Chapter 2: Basic Principles of Survival Analysis

Understanding Hazard and Survival Functions

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0.1 Learning Objectives

- Define and interpret survival and hazard functions
- Calculate survival probabilities from hazard functions
- Work with parametric survival distributions (exponential, Weibull, gamma)
- Compute mean and median survival times
- Apply maximum likelihood estimation to simple survival problems

0.2 Setup

```r
library(survival)
library(ggplot2)
library(dplyr)
library(gridExtra)

# Set random seed for reproducibility
set.seed(123)
```

0.3 The Hazard and Survival Functions

0.3.1 Key Definitions

The **survival function** defines the probability of surviving up to time t:

$$S(t) = P(T > t), \quad 0 < t < \infty$$

- Takes value 1 at time 0
- Decreases (or remains constant) over time

- Never drops below 0
- Right continuous

The **hazard function** is the instantaneous failure rate:

$$h(t) = \lim_{\delta \to 0} \frac{P(t < T < t + \delta | T > t)}{\delta}$$

- Also called the intensity function or force of mortality
- Probability of failure in the next small interval, given survival to time t

0.3.2 Key Relationships

The hazard and survival functions are related by:

$$h(t) = \frac{f(t)}{S(t)}$$

where $f(t)$ is the probability density function.

The **cumulative hazard function** is:

$$H(t) = \int_0^t h(u)du$$

And the survival function can be expressed as:

$$S(t) = \exp(-H(t))$$

0.4 Light Bulb Demonstration: Survival Analysis in Action

0.4.1 Exercise 0: Physical Demonstration (20 minutes)

Materials: 5 light bulbs with failure times pre-marked

Let's make survival analysis concrete with a physical demonstration.

0.4.1.1 Part A: Data Collection (5 minutes)

Reveal the failure times written on each bulb:

```
Bulb A: Failed at ___ hours
Bulb B: Failed at ___ hours
Bulb C: Censored at ___ hours (still working when removed)
Bulb D: Failed at ___ hours
Bulb E: Censored at ___ hours (still working when removed)
```

0.4.1.2 Part B: Manual S(t) Construction (10 minutes)

Step 1: Organize events chronologically

```
Time ___: Bulb ___ fails
Time ___: Bulb ___ fails
Time ___: Bulb ___ censored
Time ___: Bulb ___ fails
Time ___: Bulb ___ censored
```

Step 2: Build survival table by hand

Time	At Risk	Deaths	Censored	Conditional Survival	S(t)
0	5	0	0	1.0	1.0
___	___	___	___	/	___
___	___	___	___	/	___
___	___	___	___	/	___
___	___	___	___	/	___
___	___	___	___	/	___

Step 3: Draw survival curve on blackboard

- Plot time on x-axis, S(t) on y-axis
- Show step function dropping only at death times
- Mark censoring points with tick marks

0.4.1.3 Part C: R Visualization (5 minutes)

```r
# Input actual class data here
bulb_times <- c(3, 7, 10, 12, 15)  # Replace with actual times
bulb_status <- c(1, 1, 0, 1, 0)    # 1=failed, 0=censored

# Manual calculation verification
cat("Manual Calculations:\n")
```

```
Manual Calculations:
```

```r
cat("Bulb-hours:", sum(bulb_times), "\n")
```

```
Bulb-hours: 47
```

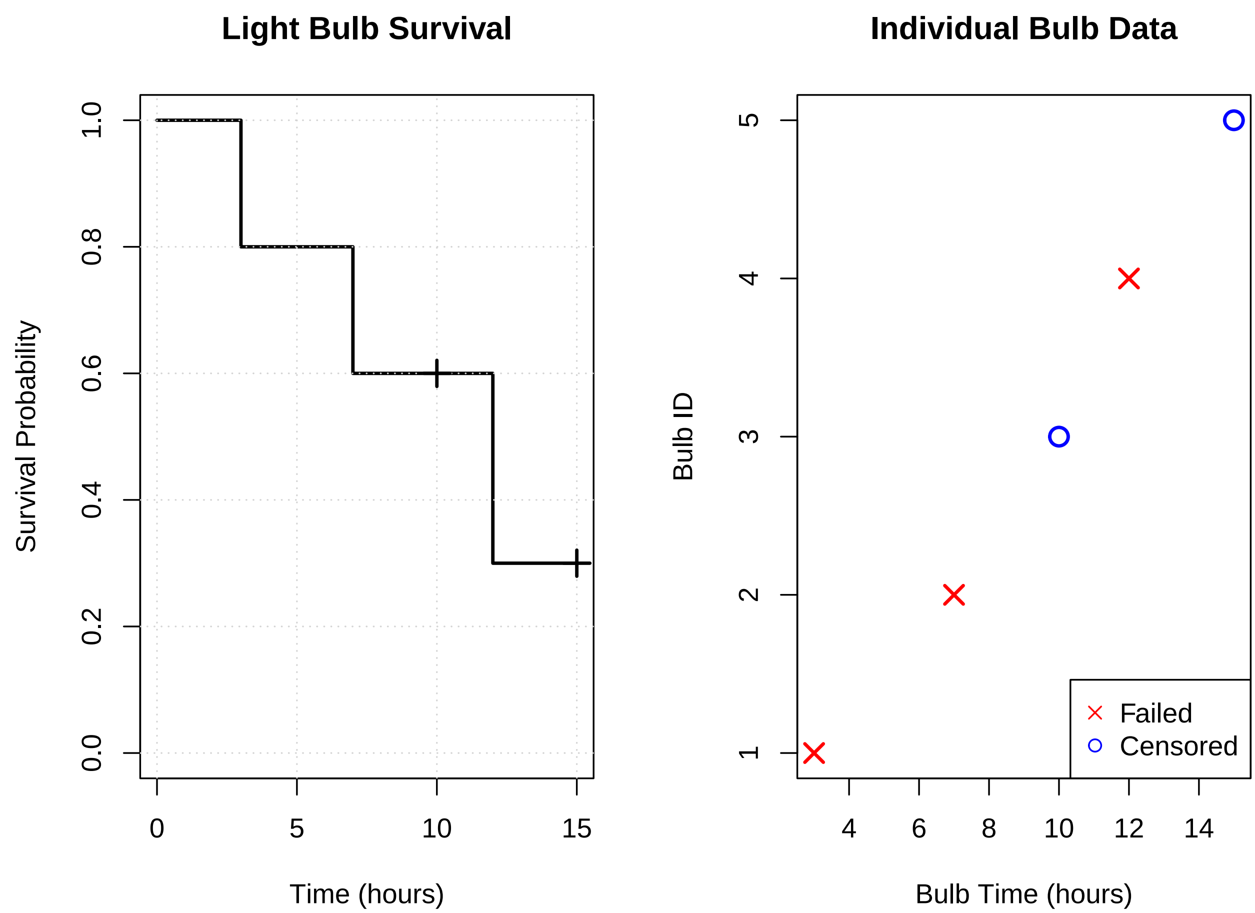
```r
cat("Failures:", sum(bulb_status), "\n")
```

```
Failures: 3
```

```r
cat("Failure rate:", round(sum(bulb_status)/sum(bulb_times), 4), "per hour\n\n")
```

```
Failure rate: 0.0638 per hour
```

```r
# Kaplan-Meier curve
bulb_surv <- Surv(bulb_times, bulb_status)
km_fit <- survfit(bulb_surv ~ 1)

# Create detailed plot
par(mfrow=c(1,2))

# Plot 1: Basic survival curve
plot(km_fit, xlab="Time (hours)", ylab="Survival Probability",
     main="Light Bulb Survival", conf.int=FALSE, lwd=2)
grid()

# Add censoring marks
censor_times <- bulb_times[bulb_status == 0]
if(length(censor_times) > 0) {
  censor_surv <- summary(km_fit, times=censor_times)$surv
  points(censor_times, censor_surv, pch=3, cex=1.5, lwd=2)
}

# Plot 2: Step-by-step data
plot(bulb_times, 1:length(bulb_times),
     pch=ifelse(bulb_status==1, 4, 1),
     col=ifelse(bulb_status==1, "red", "blue"),
     xlab="Bulb Time (hours)", ylab="Bulb ID",
     main="Individual Bulb Data",
     cex=1.5, lwd=2)
legend("bottomright", c("Failed", "Censored"),
       pch=c(4,1), col=c("red", "blue"))
```




```r
par(mfrow=c(1,1))

# Print survival table
cat("Kaplan-Meier Table:\n")
```

```
Kaplan-Meier Table:
```

```r
print(summary(km_fit))
```

```
Call: survfit(formula = bulb_surv ~ 1)

 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    3      5       1      0.8   0.179       0.5161            1
    7      4       1      0.6   0.219       0.2933            1
   12      2       1      0.3   0.239       0.0631            1
```

0.5 Building S(t) Step-by-Step: Patient Data

0.5.1 Exercise 1: Manual Construction (15 minutes)

Dataset: 6 patients - let's build the survival curve by hand

Patient	Time	Status
1	1	Death
2	2	Death
3	3	Censored
4	4	Death
5	4	Death
6	5	Censored

0.5.1.1 Step-by-Step Manual Calculation (10 minutes)

Create survival table together:

Time	At Risk	Deaths	Conditional Survival	S(t)	Calculation
0	6	0	1.0	1.0	Starting point
1	____	____	____	____	1.0 × (____ / ____)
2	____	____	____	____	____ × (____ / ____)
3	____	____	____	____	No deaths → same
4	____	____	____	____	____ × (____ / ____)
5	____	____	____	____	____ × (____ / ____)

Key Questions to Work Through: 1. At time 3, patient 3 is censored. What happens to S(t)? 2. At time 4, two patients die. How do we handle this? 3. When do we stop calculating?

0.5.1.2 Visualization in R (5 minutes)

```r
# Patient data
time <- c(1, 2, 3, 4, 4, 5)
status <- c(1, 1, 0, 1, 1, 0)

# Manual calculations first
cat("Manual Verification:\n")
```

```
Manual Verification:
```

```r
cat("Person-time:", sum(time), "years\n")
```

```
Person-time: 19 years
```

```r
cat("Deaths:", sum(status), "\n")
```

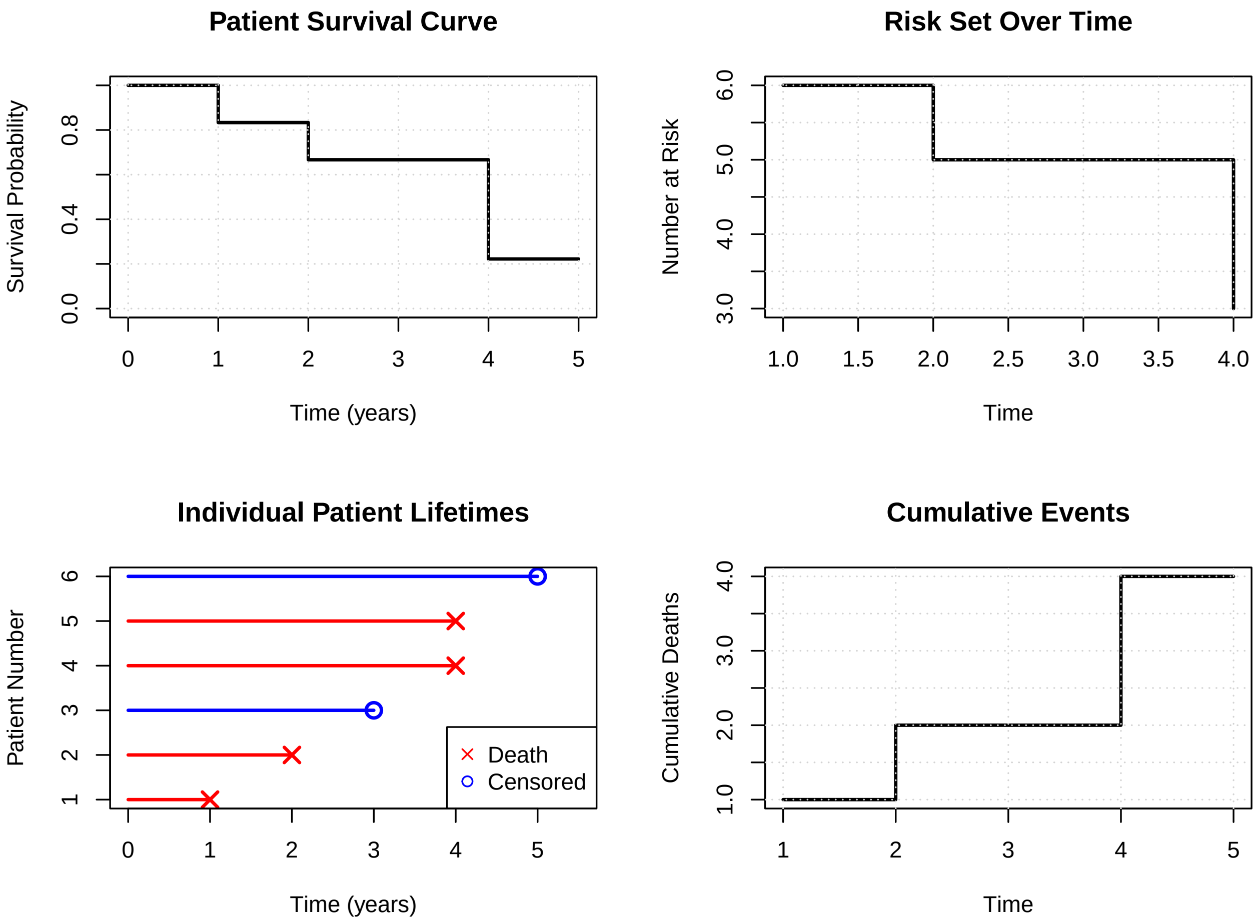
```
Deaths: 4
```

```r
cat("Overall rate:", round(sum(status)/sum(time), 4), "per year\n\n")
```

```
Overall rate: 0.2105 per year
```

```r
# Survival analysis
surv_obj <- Surv(time, status)
km_patient <- survfit(surv_obj ~ 1)

# Create comprehensive visualization
par(mfrow=c(2,2))

# Plot 1: Survival curve
plot(km_patient, xlab="Time (years)", ylab="Survival Probability",
     main="Patient Survival Curve", conf.int=FALSE, lwd=2)
grid()

# Plot 2: Risk set over time
times <- sort(unique(time[status==1]))
risk_set <- sapply(times, function(t) sum(time >= t))
plot(times, risk_set, type="s", lwd=2,
     xlab="Time", ylab="Number at Risk",
     main="Risk Set Over Time")
grid()

# Plot 3: Individual patient data (dumbbell timeline)
plot(time, 1:6, pch=ifelse(status==1, 4, 1),
     col=ifelse(status==1, "red", "blue"),
     xlab="Time (years)", ylab="Patient Number",
     main="Individual Patient Lifetimes",
     cex=1.5, lwd=2, xlim=c(0, max(time)*1.1))

# Add horizontal lines from 0 to each event time
for(i in 1:6) {
  segments(0, i, time[i], i,
           col=ifelse(status[i]==1, "red", "blue"),
```

```r
              lwd=2)
}

legend("bottomright", c("Death", "Censored"),
       pch=c(4,1), col=c("red", "blue"))

# Plot 4: Cumulative events
cum_events <- cumsum(status[order(time)])
ordered_times <- sort(time)
plot(ordered_times, cum_events, type="s", lwd=2,
     xlab="Time", ylab="Cumulative Deaths",
     main="Cumulative Events")
grid()
```




```r
par(mfrow=c(1,1))

# Detailed survival table
cat("Detailed Survival Table:\n")
```

```
Detailed Survival Table:
```

```r
print(summary(km_patient))
```

```
Call: survfit(formula = surv_obj ~ 1)

 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    1      6       1    0.833   0.152       0.5827            1
    2      5       1    0.667   0.192       0.3786            1
    4      3       2    0.222   0.192       0.0407            1
```

0.6 Exponential Model: Theory Meets Practice

0.6.1 Exercise 2: By-Hand Exponential Calculations (10 minutes)

Using our patient data, let's fit an exponential model manually.

0.6.1.1 MLE Calculation by Hand

```
ˆ = Deaths ÷ Person-time = ___ ÷ ___ = ___ per year
```

0.6.1.2 Exponential Predictions

```
Mean survival = 1/ˆ = ___
Median survival = ln(2)/ˆ = ___
S(3) = exp(-ˆ × 3) = ___
```

0.6.1.3 Compare Models Visually

```r
# Calculate MLE
d <- sum(status)
V <- sum(time)
lambda_hat <- d/V

cat("Exponential Model:\n")
```

```
Exponential Model:
```

```r
cat("ˆ =", round(lambda_hat, 4), "per year\n")
```

```
ˆ = 0.2105 per year
```

```r
cat("Mean survival =", round(1/lambda_hat, 2), "years\n")
```

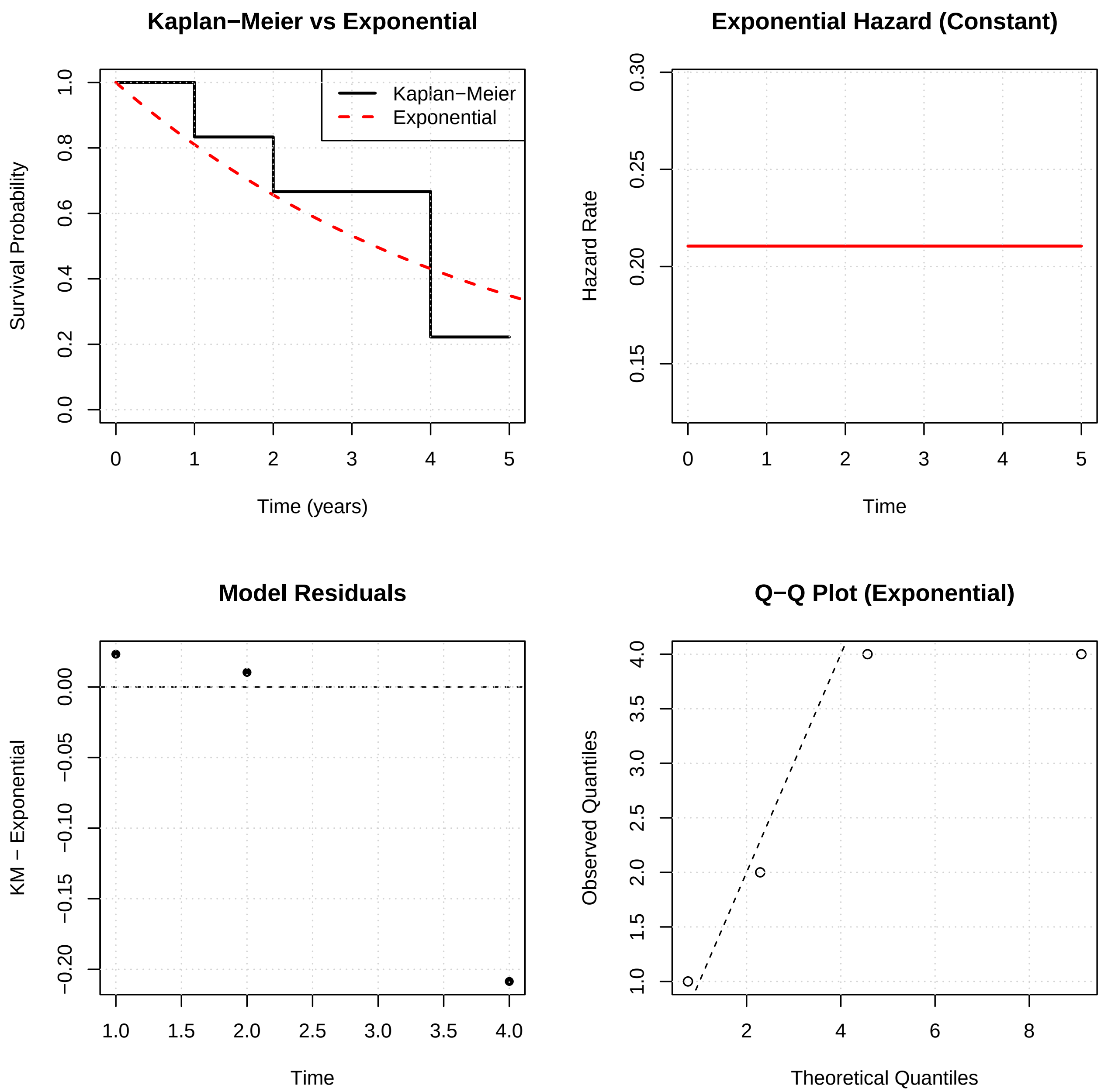
```
Mean survival = 4.75 years
```

```r
cat("Median survival =", round(log(2)/lambda_hat, 2), "years\n\n")
```

```
Median survival = 3.29 years
```

```r
# Create comparison plots
par(mfrow=c(2,2))

# Plot 1: Both survival curves
plot(km_patient, conf.int=FALSE, lwd=2,
     xlab="Time (years)", ylab="Survival Probability",
     main="Kaplan-Meier vs Exponential")
curve(exp(-lambda_hat * x), from=0, to=6, add=TRUE,
      col="red", lwd=2, lty=2)
legend("topright", c("Kaplan-Meier", "Exponential"),
       col=c("black", "red"), lwd=2, lty=c(1,2))
grid()

# Plot 2: Hazard functions
plot(c(0, max(time)), c(lambda_hat, lambda_hat), type="l",
     lwd=2, col="red", xlab="Time", ylab="Hazard Rate",
     main="Exponential Hazard (Constant)")
grid()

# Plot 3: Residuals
km_times <- summary(km_patient)$time
km_surv <- summary(km_patient)$surv
exp_surv <- exp(-lambda_hat * km_times)
plot(km_times, km_surv - exp_surv, type="p", pch=16,
     xlab="Time", ylab="KM - Exponential",
     main="Model Residuals")
abline(h=0, lty=2)
grid()

# Plot 4: Q-Q plot for exponential
theoretical_quantiles <- qexp(ppoints(length(time[status==1])), rate=lambda_hat)
observed_quantiles <- sort(time[status==1])
plot(theoretical_quantiles, observed_quantiles,
     xlab="Theoretical Quantiles", ylab="Observed Quantiles",
     main="Q-Q Plot (Exponential)")
abline(0, 1, lty=2)
```

```r
grid()
```




```r
par(mfrow=c(1,1))

# Summary comparison
cat("Model Comparison at t=3:\n")
```

```
Model Comparison at t=3:
```

```r
km_3 <- summary(km_patient, times=3)$surv
exp_3 <- exp(-lambda_hat * 3)
cat("Kaplan-Meier S(3) =", round(km_3, 3), "\n")
```

```
Kaplan-Meier S(3) = 0.667
```

```r
cat("Exponential S(3) =", round(exp_3, 3), "\n")
```

```
Exponential S(3) = 0.532
```

```r
cat("Difference =", round(km_3 - exp_3, 3), "\n")
```

```
Difference = 0.135
```

Let's work through a small dataset by hand. Consider these 6 patients:

Patient	Time	Status
1	7	Death
2	6	Death
3	6	Censored
4	5	Death
5	2	Censored
6	4	Censored

0.6.2 Exercise 1A: Basic Calculations (Do by Hand)

Step 1: Calculate the total person-time at risk

```
Person-time = 7 + 6 + 6 + 5 + 2 + 4 = ___ years
```

Step 2: Count the total number of deaths

```
Deaths = ___
```

Step 3: Calculate the overall hazard rate (deaths/person-time)

```
Hazard rate = Deaths / Person-time = ___ / ___ = ___
```

0.6.3 Exercise 1B: Verify with R

```r
# Patient data
time <- c(7, 6, 6, 5, 2, 4)
status <- c(1, 1, 0, 1, 0, 0)  # 1 = death, 0 = censored

# Calculate person-time and deaths
person_time <- sum(time)
deaths <- sum(status)

print(paste("Person-time:", person_time))
```

```
[1] "Person-time: 30"
```

```r
print(paste("Deaths:", deaths))
```

```
[1] "Deaths: 3"
```

```r
print(paste("Hazard rate:", deaths/person_time))
```

```
[1] "Hazard rate: 0.1"
```

0.7 Parametric Survival Distributions

0.7.1 The Exponential Distribution

The simplest survival distribution has a **constant hazard**: $h(t) = \lambda$

Key Properties:

- Hazard: $h(t) = \lambda$ (constant)
- Cumulative hazard: $H(t) = \lambda t$
- Survival: $S(t) = e^{-\lambda t}$
- Mean survival: $E(T) = 1/\lambda$
- Median survival: $t_{med} = \ln(2)/\lambda$

0.7.2 Exercise 2: Exponential Distribution Calculations (By Hand)

Suppose we have an exponential distribution with $\lambda = 0.1$ (per year).

Step 1: Calculate the probability of surviving beyond 5 years

```
S(5) = e^(- t) = e^(-0.1 × 5) = e^(-0.5) = ___
```

Step 2: Calculate the mean survival time

```
E(T) = 1/  = 1/0.1 = ___ years
```

Step 3: Calculate the median survival time

```
t_med = ln(2)/  = 0.693/0.1 = ___ years
```

Step 4: What's the hazard of death at any time t?

```
h(t) = ___ (constant)
```

0.7.3 Verify Your Calculations

```r
lambda <- 0.1

# Survival at 5 years
surv_5 <- exp(-lambda * 5)
print(paste("S(5) =", round(surv_5, 4)))
```

```
[1] "S(5) = 0.6065"
```

```r
# Mean survival
mean_surv <- 1/lambda
print(paste("Mean survival =", mean_surv, "years"))
```

```
[1] "Mean survival = 10 years"
```

```r
# Median survival
median_surv <- log(2)/lambda
print(paste("Median survival =", round(median_surv, 2), "years"))
```

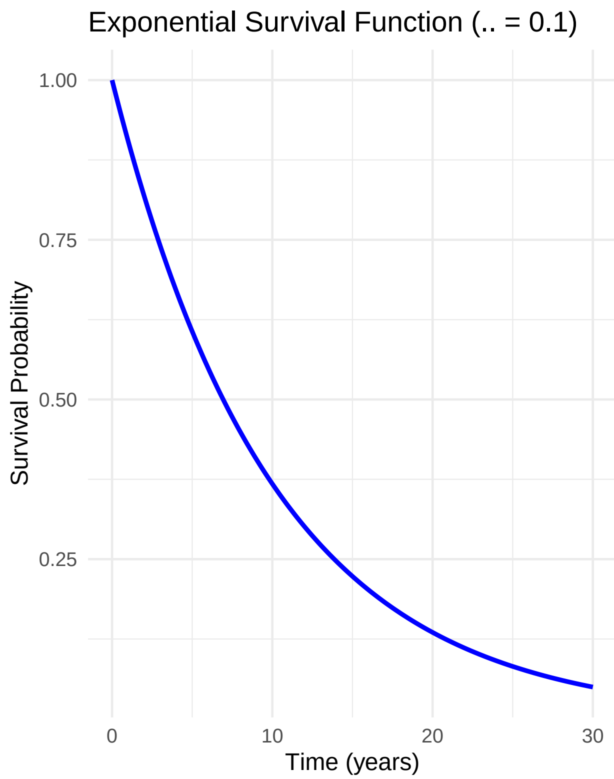
```
[1] "Median survival = 6.93 years"
```

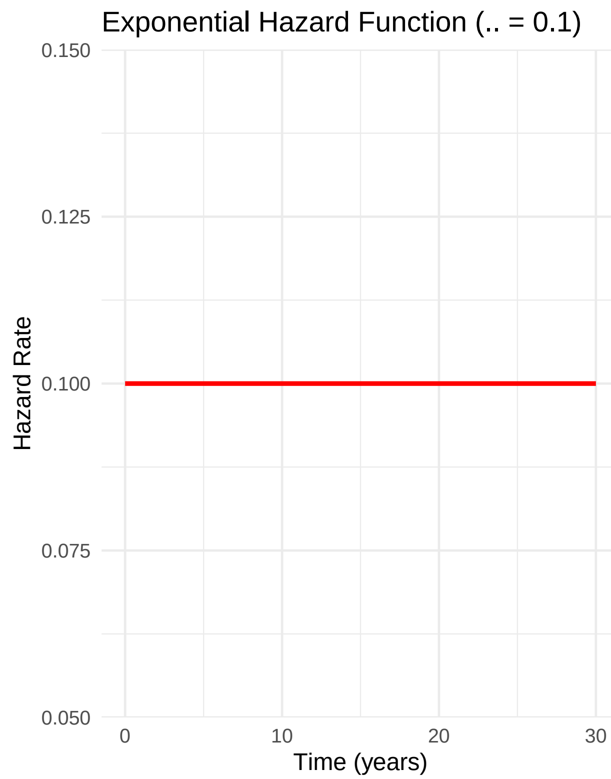
```r
# Verify: S(median) should equal 0.5
print(paste("S(median) =", round(exp(-lambda * median_surv), 3)))
```

```
[1] "S(median) = 0.5"
```

0.7.4 Visualizing the Exponential Distribution

```r
# Create time sequence
time_seq <- seq(0, 30, 0.1)

# Calculate survival and hazard functions
surv_func <- exp(-lambda * time_seq)
hazard_func <- rep(lambda, length(time_seq))

# Create plots
p1 <- ggplot(data.frame(time = time_seq, survival = surv_func),
             aes(x = time, y = survival)) +
  geom_line(size = 1, color = "blue") +
  labs(title = "Exponential Survival Function (  = 0.1)",
       x = "Time (years)", y = "Survival Probability") +
  theme_minimal()

p2 <- ggplot(data.frame(time = time_seq, hazard = hazard_func),
             aes(x = time, y = hazard)) +
  geom_line(size = 1, color = "red") +
  labs(title = "Exponential Hazard Function (  = 0.1)",
       x = "Time (years)", y = "Hazard Rate") +
  theme_minimal()

grid.arrange(p1, p2, ncol = 2)
```





0.8 The Weibull Distribution

More flexible than exponential, with hazard function:

$$h(t) = \alpha\lambda^{\alpha}t^{\alpha-1}$$

Parameters:

- α = shape parameter
- λ = scale parameter

Key Properties:

- When $\alpha = 1$: reduces to exponential
- When $\alpha > 1$: increasing hazard
- When $\alpha < 1$: decreasing hazard

0.8.1 Exercise 3: Weibull Distribution (By Hand)

Consider a Weibull distribution with $\alpha = 2$ and $\lambda = 0.1$.

Step 1: What type of hazard pattern does this represent?

```
Since   = 2 > 1, the hazard is _______ over time
```

Step 2: Calculate the hazard at time t = 5 years

```
h(5) =   ^  t^( -1) = 2 × (0.1)^2 × 5^(2-1) = 2 × 0.01 × 5 = ___
```

Step 3: Calculate the cumulative hazard at t = 5

```
H(5) = ( t)^  = (0.1 × 5)^2 = (0.5)^2 = ___
```

Step 4: Calculate the survival probability at t = 5

```
S(5) = exp(-H(5)) = exp(-0.25) = ___
```

0.8.2 Verify Weibull Calculations

```r
alpha <- 2
lambda <- 0.1
t <- 5

# Hazard at t=5
hazard_5 <- alpha * (lambda^alpha) * (t^(alpha-1))
print(paste("h(5) =", hazard_5))
```

```
[1] "h(5) = 0.1"
```

```r
# Cumulative hazard at t=5
cum_hazard_5 <- (lambda * t)^alpha
print(paste("H(5) =", cum_hazard_5))
```

```
[1] "H(5) = 0.25"
```

```r
# Survival at t=5
surv_5_weibull <- exp(-cum_hazard_5)
print(paste("S(5) =", round(surv_5_weibull, 4)))
```

```
[1] "S(5) = 0.7788"
```

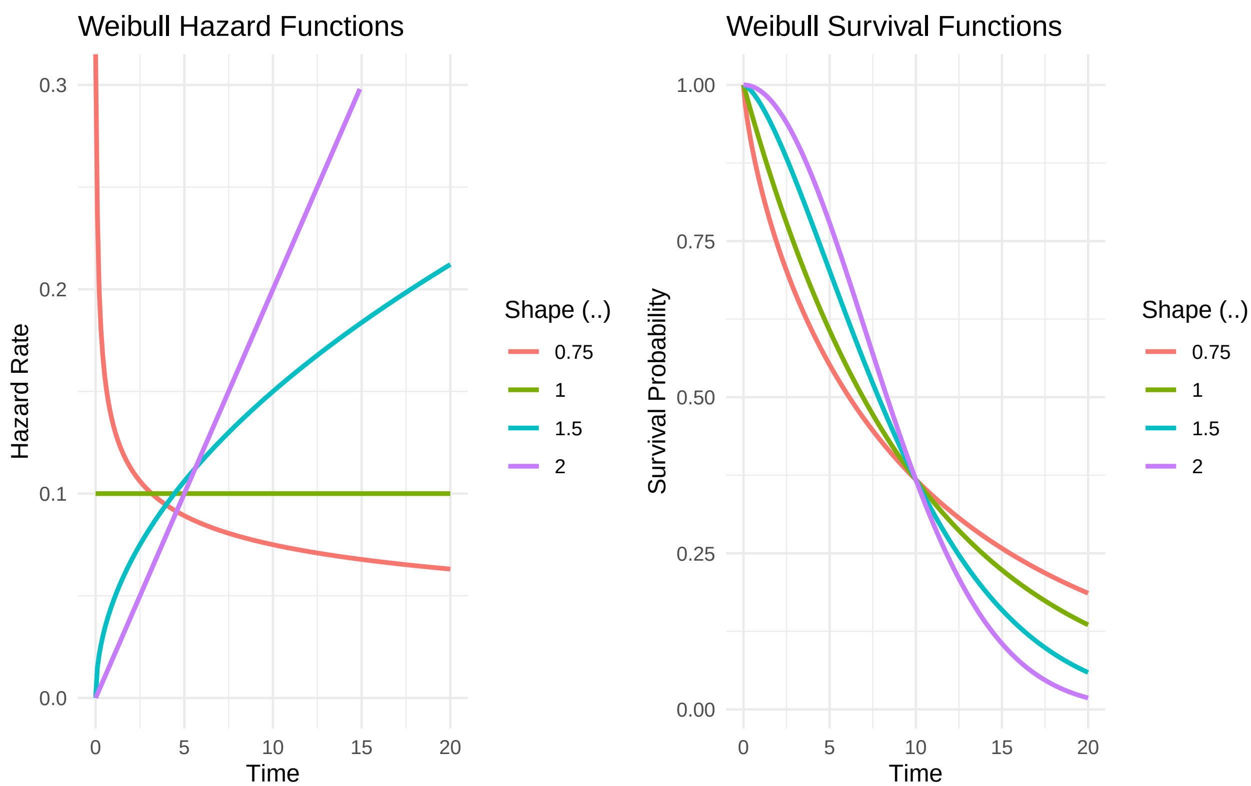
```r
# Verify using R's built-in function
surv_5_r <- pweibull(5, shape=alpha, scale=1/lambda, lower.tail=FALSE)
print(paste("S(5) using R =", round(surv_5_r, 4)))
```

```
[1] "S(5) using R = 0.7788"
```

0.8.3 Comparing Different Weibull Shapes

```r
# Define parameters for comparison
time_seq <- seq(0, 20, 0.1)
lambda <- 0.1

# Different shape parameters
alphas <- c(0.75, 1, 1.5, 2)

# Create data frame for plotting
weibull_data <- data.frame()

for(alpha in alphas) {
  hazard <- alpha * (lambda^alpha) * (time_seq^(alpha-1))
  survival <- exp(-(lambda * time_seq)^alpha)

  temp_data <- data.frame(
    time = time_seq,
    hazard = hazard,
    survival = survival,
    alpha = as.factor(alpha)
  )
  weibull_data <- rbind(weibull_data, temp_data)
}

# Plot hazard functions
p1 <- ggplot(weibull_data, aes(x = time, y = hazard, color = alpha)) +
  geom_line(size = 1) +
  labs(title = "Weibull Hazard Functions",
       x = "Time", y = "Hazard Rate",
       color = "Shape ( )") +
  theme_minimal() +
  ylim(0, 0.3)

# Plot survival functions
p2 <- ggplot(weibull_data, aes(x = time, y = survival, color = alpha)) +
  geom_line(size = 1) +
  labs(title = "Weibull Survival Functions",
       x = "Time", y = "Survival Probability",
```

```
        color = "Shape ( )") +
  theme_minimal()

grid.arrange(p1, p2, ncol = 2)
```



0.9 Maximum Likelihood Estimation

0.9.1 The Likelihood Function

For survival data with censoring, the likelihood is:

$$L(\theta) = \prod_{i=1}^{n} h(t_i; \theta)^{\delta_i} S(t_i; \theta)$$

where $\delta_i = 1$ for deaths and $\delta_i = 0$ for censored observations.

0.9.2 Exercise 4: MLE for Exponential Distribution (By Hand)

Using our 6-patient dataset from Exercise 1:

The log-likelihood for exponential distribution is:

$$\ell(\lambda) = d\log(\lambda) - \lambda V$$

where d = number of deaths and V = total person-time.

Step 1: Write the log-likelihood for our data

```
d = ___ deaths
V = ___ person-years
 ( ) = ___ log( ) -   × ___
```

Step 2: Find the MLE by taking the derivative and setting to zero

```
d /d  = d/  - V = 0
Solving: ˆ = d/V = ___ / ___ = ___
```

Step 3: Calculate the variance of the estimate

```
Var(ˆ)  ˆ²/d = (_____)²/___ = ___
Standard Error = √Var(ˆ) = ___
```

0.9.3 Verify MLE Calculation

```r
# Our data
time <- c(7, 6, 6, 5, 2, 4)
status <- c(1, 1, 0, 1, 0, 0)

d <- sum(status)  # deaths
V <- sum(time)    # person-time

# MLE
lambda_hat <- d/V
print(paste("ˆ =", round(lambda_hat, 4)))
```

```
[1] "ˆ = 0.1"
```

```r
# Variance and standard error
var_lambda <- lambda_hat^2 / d
se_lambda <- sqrt(var_lambda)

print(paste("Variance of ˆ =", round(var_lambda, 6)))
```

```
[1] "Variance of ˆ = 0.003333"
```

```r
print(paste("Standard Error =", round(se_lambda, 4)))
```

```
[1] "Standard Error = 0.0577"
```

```r
# 95% Confidence interval (using normal approximation)
ci_lower <- lambda_hat - 1.96 * se_lambda
ci_upper <- lambda_hat + 1.96 * se_lambda

print(paste("95% CI: (", round(ci_lower, 4), ",", round(ci_upper, 4), ")"))
```

```
[1] "95% CI: ( -0.0132 , 0.2132 )"
```

0.10 Hands-On Workshop Exercise

0.10.1 Exercise 5: Complete Analysis by Hand, Then Verify

You are given this survival dataset:

Patient	Time	Status
A	12	Death
B	8	Death
C	15	Censored
D	10	Death
E	6	Censored
F	9	Death

Part A: Basic Statistics (Work by hand first)

1. Total person-time: ________________
2. Number of deaths: ________________
3. MLE for exponential : ________________
4. Mean survival time: ________________
5. Median survival time: ________________

Part B: Probability Calculations

Assuming exponential distribution with your estimated :

1. P(survive > 10 years) = ________________
2. P(die between years 5-10) = ________________
3. Hazard rate at any time = ________________

Part C: Verify Your Work

```
# Fill in the values
time_workshop <- c(__, __, __, __, __, __)
status_workshop <- c(__, __, __, __, __, __)  # 1=death, 0=censored

# Calculate basic statistics
person_time_total <- sum(time_workshop)
deaths_total <- sum(status_workshop)
lambda_est <- deaths_total / person_time_total

print(paste("Person-time:", person_time_total))
print(paste("Deaths:", deaths_total))
print(paste("  estimate:", round(lambda_est, 4)))
print(paste("Mean survival:", round(1/lambda_est, 2)))
print(paste("Median survival:", round(log(2)/lambda_est, 2)))

# Probability calculations
prob_survive_10 <- exp(-lambda_est * 10)
prob_die_5_to_10 <- exp(-lambda_est * 5) - exp(-lambda_est * 10)

print(paste("P(survive > 10):", round(prob_survive_10, 4)))
print(paste("P(die between 5-10):", round(prob_die_5_to_10, 4)))
```

0.10.2 Solution

```
# Solution
time_workshop <- c(12, 8, 15, 10, 6, 9)
status_workshop <- c(1, 1, 0, 1, 0, 1)

person_time_total <- sum(time_workshop)
deaths_total <- sum(status_workshop)
lambda_est <- deaths_total / person_time_total

print("=== SOLUTION ===")
```

```
[1] "=== SOLUTION ==="
```

```
print(paste("Person-time:", person_time_total, "years"))
```

```
[1] "Person-time: 60 years"
```

```
print(paste("Deaths:", deaths_total))
```

```
[1] "Deaths: 4"
```

```r
print(paste("  estimate:", round(lambda_est, 4)))
```

```
[1] "  estimate: 0.0667"
```

```r
print(paste("Mean survival:", round(1/lambda_est, 2), "years"))
```

```
[1] "Mean survival: 15 years"
```

```r
print(paste("Median survival:", round(log(2)/lambda_est, 2), "years"))
```

```
[1] "Median survival: 10.4 years"
```

```r
# Probability calculations
prob_survive_10 <- exp(-lambda_est * 10)
prob_die_5_to_10 <- exp(-lambda_est * 5) - exp(-lambda_est * 10)

print(paste("P(survive > 10 years):", round(prob_survive_10, 4)))
```

```
[1] "P(survive > 10 years): 0.5134"
```

```r
print(paste("P(die between 5-10 years):", round(prob_die_5_to_10, 4)))
```

```
[1] "P(die between 5-10 years): 0.2031"
```

```r
print(paste("Hazard rate:", round(lambda_est, 4), "(constant)"))
```

```
[1] "Hazard rate: 0.0667 (constant)"
```

0.11 Summary and Key Takeaways

0.11.1 Essential Formulas to Remember

Exponential Distribution: - Survival: $S(t) = e^{-\lambda t}$
- Hazard: $h(t) = \lambda$ (constant) - Mean: $1/\lambda$ - Median: $\ln(2)/\lambda$ - MLE: $\hat{\lambda} = d/V$

General Relationships: - $S(t) = \exp(-H(t))$ where $H(t) = \int_0^t h(u)du$ - $h(t) = f(t)/S(t)$ - Mean survival $= \int_0^\infty S(t)dt$

0.12 Additional Practice Problems

0.12.1 Problem 1

If the hazard rate is constant at 0.05 per year, what percentage of patients survive beyond 20 years?

0.12.2 Problem 2

A study reports a median survival of 8 years. Assuming exponential distribution, what is the hazard rate?

0.12.3 Problem 3

Given survival times: 3, 5, 7+, 9, 12+ (+ indicates censoring), calculate the MLE for .

0.13 References

Moore, D. *Applied Survival Analysis Using R.* Chapter 2: Basic Principles of Survival Analysis.

0.14 Session Information

```r
sessionInfo()
```

```
R version 4.3.1 (2023-06-16)
Platform: x86_64-apple-darwin20 (64-bit)
Running under: macOS 15.6

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/lib/libRlapack.dylib;  L

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/New_York
tzcode source: internal

attached base packages:
```

```
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] gridExtra_2.3  dplyr_1.1.4    ggplot2_3.5.2  survival_3.8-3

loaded via a namespace (and not attached):
 [1] vctrs_0.6.5       cli_3.6.3         knitr_1.44        rlang_1.1.4
 [5] xfun_0.52         generics_0.1.3    jsonlite_1.8.7    labeling_0.4.3
 [9] glue_1.8.0        colorspace_2.1-0  htmltools_0.5.8.1 fansi_1.0.6
[13] scales_1.3.0      rmarkdown_2.25    grid_4.3.1        evaluate_0.22
[17] munsell_0.5.1     tibble_3.2.1      fastmap_1.1.1     yaml_2.3.7
[21] lifecycle_1.0.4   compiler_4.3.1    pkgconfig_2.0.3   rstudioapi_0.15.0
[25] farver_2.1.2      lattice_0.21-8    digest_0.6.33     R6_2.5.1
[29] tidyselect_1.2.1  utf8_1.2.4        pillar_1.9.0      splines_4.3.1
[33] magrittr_2.0.3    Matrix_1.6-5      withr_3.0.2       tools_4.3.1
[37] gtable_0.3.6
```

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